

Estimating AIDS and QUAIDS demand systems with survey data: the equaids Stata module

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Abstract

`equaids` estimates the almost ideal demand system of Deaton and Muellbauer (1980) and its quadratic extension by Banks et al. (1997), with demographic variables entering by the scaling of Ray (1983), by iterated feasible generalized nonlinear least squares. The estimator is the one of Poi (2012) and of Stata's `demandsys`; `equaids` computes it in Mata with an analytic Jacobian, stops on a criterion with a statistical meaning (the estimate lies within 0.001 standard errors of the optimum), and adds what earlier programs lack: summary elasticities defined as means over the households (of the households by default, of the individuals, or of the market, the elasticity of total demand), with analytic standard errors that include the sampling of the households and are the exact derivative of the whole procedure; variances robust, by cluster or by survey design (`svyset`: strata, primary sampling units, finite-population correction), all built from the same influence functions; and diagnostics of the data and of the estimate. The design variance equals Stata's own linearization to 4×10^{-15} , and the standard errors of 91 market elasticities agree with the bootstrap of the households and with the Rao–Wu bootstrap of a survey design, within the Monte Carlo error. In a Monte Carlo on synthetic populations built to satisfy the theory and to identify every parameter, the 95% intervals cover 0.945 to 0.955 under simple random sampling and 0.932 to 0.948 under a stratified cluster design with 200 sampling units; a brute-force influence function (the whole procedure re-run with the weight of each household moved) reproduces the robust standard errors within 2×10^{-4} . A diagnostic command, `equaidsdiag`, checks a specification before it is estimated, and measures how the elasticities move when a demographic variable is left out.

Keywords: demand systems; AIDS; QUAIDS; aggregate elasticities; survey design; influence functions; Stata.

JEL classification: C13, C83, C87, D12.

*E-mail: aabd@ecn.ulaval.ca. The module described here, `equaids` 1.2.0, is available at <https://github.com/aabdd12/equaids> (Section 8.1). This note: <https://doi.org/10.5281/zenodo.22959991>.

1 Introduction

The almost ideal demand system (AIDS) of [Deaton and Muellbauer \(1980\)](#) and its quadratic extension (QUAIDS) by [Banks et al. \(1997\)](#) are the workhorses of empirical demand analysis on household surveys: budget shares linear (AIDS) or quadratic (QUAIDS) in the logarithm of deflated expenditure, prices entering through a translog cost index, adding-up, homogeneity and symmetry imposed by linear restrictions. Their estimation in Stata has a history of about twenty years, summarized in [Table 1](#).

Table 1: Programs for AIDS and QUAIDS in Stata

Program	Reference	Estimator	Elasticities and their standard errors
evaluator for <code>nlsur</code> <code>quaid</code> s 1.1.0	Poi (2002, 2008) Poi (2012)	written by the user <code>nlsur</code> , iterated FGNLS; AIDS, QUAIDS; Ray scaling	written by the user closed forms; delta method with a numerical Jacobian, at the means or for one observation; no weights
<code>aidsills</code>	Lecocq and Robin (2015)	iterated linear least squares (Blundell and Robin, 1999); instruments for prices and expenditure	at weighted means, predicted shares; finite-difference Jacobian; fweights and aweights only
<code>wquaid</code> s (WELCOM) <code>quaid</code> sce 1.1.1 (SSC)	Araar et al. (2018) Caro, Melo, Molina and Salgado	calls Poi (2012) with analytic weights Poi (2012) plus the two-step censoring of Shonkwiler and Yen (1999) , every good corrected, no adding-up; no weights	at unweighted means, no standard errors at the means, delta method; the probits held fixed in the variance of the system
<code>demandsys</code>	StataCorp (2023)	<code>nlsur</code> , iterated FGNLS; eight systems	household elasticities at predicted shares, averaged (unweighted) or at the means; delta method

Each program solved part of the problem. [Poi \(2012\)](#) made the QUAIDS a one-line command, with Ray’s demographic scaling and closed-form elasticities. [Lecocq and Robin \(2015\)](#) brought the iterated linear least squares of [Blundell and Robin \(1999\)](#) and, with it, instruments for endogenous expenditure and prices. `demandsys` (Stata 18) generalized the estimation to eight systems and to sampling weights with a robust or clustered variance. Three gaps remain, and they matter most for the use of these systems on household surveys, which is their main use.

The survey design. None of these programs accepts the `svy` prefix, so none gives standard errors that account for strata, primary sampling units and the finite-population correction.

The summary elasticity. A demand system gives an elasticity for every household. To report one number per good, the programs either evaluate the elasticities at the means of the regressors or average the household elasticities. Both are awkward on survey data. The first describes a household that need not exist; the second divides, for every household, by its predicted budget share, which QUAIDS can predict close to zero or negative for small goods, so that a few households drive the mean ([Section 4](#)). Neither is the elasticity of the demand of the population, which is what a price or tax simulation needs. And the standard errors treat the evaluation point as fixed, ignoring that it is itself estimated from the sample.

The numerics. The `nlsur`-based programs differentiate the model numerically at every iteration, and the choice of the price-index constant α_0 , fixed by the user, can make the estimation fragile ([Section 2.1](#)).

equaids addresses the three. It estimates the same model as [Poi \(2012\)](#) and **demandsys** by the same estimator, and reproduces their estimates (Section 9); it adds summary elasticities that are means over the households (Section 4.1), analytic standard errors for them that include the sampling of the households, one variance machinery for robust, cluster and survey-design inference, and diagnostics. Version 1.1 adds the filling of missing prices and the correction for the non-buyers of [Shonkwiler and Yen \(1999\)](#) (Section 6); version 1.2 makes the mean over the households the default summary, as in **easi** and **duvm**, and computes the standard errors as the exact derivative of the procedure (Section 5). Instrumental variables (as in [Lecocq and Robin, 2015](#)) are left for later versions; beyond the rank-three Engel curves of QUAIDS, the EASI system of [Lewbel and Pendakur \(2009\)](#) is the natural next step.

The table below sets what **equaids** adds against the programs of Table 1 (for **quaidisce**, as read in its code, version 1.1.1), element by element: a method of its own (*new*), a known method brought to demand-system programs (*new here*), or a method used as it is (*as is*).

What equaids adds

Element	Earlier programs	equaids	Status
<i>Model and estimator</i>			
AIDS, QUAIDS, Ray's scaling, iterated FGNLS	Poi (2012) , demandsys	the same model and estimator, the same estimates (Section 9)	as is
Non-buyers	quaidsc : Shonkwiler and Yen (1999) on every good, no adding-up	Shonkwiler and Yen (1999) with a good bought by all closing the budget (adding-up kept)	as is, new here
Probit of purchase	quaidsc : $\ln p$ and $\ln x$ separately	on $\ln(p/x)$: expected shares homogeneous, aggregation identities exact (Section 6)	new
Goods corrected, exclusions	quaidsc : every good, no exclusion variable	chosen goods (selgoods()), variables of the probits common or specific to a good (selvars())	new here
<i>Computation and calibration</i>			
Algorithm	nlsur , numerical derivatives	Gauss-Newton in Mata, analytic Jacobian, 15 to 21 times faster (Section 9.3)	new here
α_0	set by the user	default rule, and the diagnosis of its link with the identification of λ (Section 2.1)	new
Missing prices	households lost	group means, fallback levels (pimpute())	new here
<i>Inference</i>			
Coefficients	demandsys , Poi: sandwich with the Gauss-Newton information	the exact derivative of the procedure: observed Jacobian, estimation of Σ , observed Hessian of the probits, imputation of the prices (Section 5)	new here
Summary elasticity	at the means, or mean of the household elasticities	means over the households weighted three ways (households by default, individuals, market), their standard errors including the sampling of the households (Section 4.1)	new
Other types	delta method at a fixed point	influence functions including the point of evaluation (reference household, household mean)	new here
Non-buyers	quaidsc : probits held fixed, their variance appended	influence functions of the probits stacked into those of the system	new here
Weights, design	demandsys : weights, robust, cluster; the others: few or none	pweights , cluster, survey design (strata, PSUs, fpc)	new here
Bootstrap	quaidsc : the system only	the whole procedure on each replication	new here
<i>Diagnostics and validation</i>			
Before estimation	none	sections D0 to D7, among them the stability of the elasticities to each demographic (Section 7.3)	new
Ray's scaling	none	distance of the estimate to the boundary $m_0(z) > 0$ in standard errors; the correction pushes ρ towards it	new
Validation	none	every standard error checked against an exact oracle, a brute-force influence function, the bootstrap and a Monte Carlo (Section 9)	new here

2 The model

For household h with log prices $\ln p_h$ (M goods), total expenditure x_h and demographic variables z_h (K of them), the budget share of good i is

$$w_{ih} = \alpha_i + \sum_{j=1}^M \gamma_{ij} \ln p_{jh} + (\beta_i + \eta_i' z_h) \ell_h + \frac{\lambda_i}{b(p_h) c(p_h, z_h)} \ell_h^2 + u_{ih}, \quad \ell_h = \ln x_h - \ln m_0(z_h) - \ln a(p_h), \quad (1)$$

with the translog price index $\ln a(p) = \alpha_0 + \sum_k \alpha_k \ln p_k + \frac{1}{2} \sum_k \sum_l \gamma_{kl} \ln p_k \ln p_l$, $b(p) = \prod_k p_k^{\beta_k}$, Ray's scaling $m_0(z) = 1 + \rho'z$ and $c(p, z) = \prod_k p_k^{\eta_k' z}$ (Ray, 1983; Poi, 2012). AIDS is the case $\lambda = 0$. Adding-up, homogeneity and symmetry,

$$\sum_i \alpha_i = 1, \quad \sum_i \beta_i = \sum_i \lambda_i = 0, \quad \sum_i \eta_i = 0, \quad \sum_i \gamma_{ij} = 0, \quad \gamma_{ij} = \gamma_{ji},$$

are imposed by writing the full parameter vector as an affine function of a vector θ of free parameters (Poi, 2012); the residuals of the M equations sum to zero, and the system is estimated on $M - 1$ of them.

2.1 The constant α_0

The constant of the price index is not identified with the other parameters and is fixed. Banks et al. (1997) and Araar (1998, chap. 5) set it just below the smallest log expenditure, so that deflated expenditure ℓ is positive; Poi (2012) leaves it to the user, and his example uses $\alpha_0 = 10$, above the log expenditure of every household in his data. **equaids** sets, by default,

$$\alpha_0 = \min_h \ln x_h - 0.1$$

on the estimation sample, and **anot(#)** imposes a value (to reproduce Poi's results, for instance). The default is a rule of the data: a bootstrap or a simulation must apply it again to each sample, and **equaids**'s validation does so.

The choice matters for inference, not only for the fit. With Poi's $\alpha_0 = 10$ on his own data, the information matrix of the QUAIDS is nearly singular (scaled reciprocal condition number 3×10^{-6} , against 2×10^{-4} with the default rule and 10^{-4} to 3×10^{-3} with α_0 inside the range of $\ln x$), and the estimator is far from its normal approximation: the bootstrap standard deviation of the coefficients is up to three times the sandwich standard error (Section 9). With the default rule, the same data give standard errors that agree with the bootstrap.

3 Estimation

3.1 Iterated FGNLS

Let $f(\theta; p_h, x_h, z_h)$ be the $M - 1$ fitted shares, $G_h = \partial f / \partial \theta'$ their Jacobian, ω_h the weight of household h and Σ the covariance of the residuals. Iterated FGNLS solves

$$\sum_h \omega_h G_h' \Sigma^{-1} u_h(\theta) = 0, \quad \Sigma = \frac{\sum_h \omega_h u_h u_h'}{\sum_h \omega_h}, \quad (2)$$

the Gaussian quasi-maximum-likelihood estimator of the system, consistent whenever $\mathbb{E}[u_h \mid p_h, x_h, z_h] = 0$ and invariant to the equation left out. **equaids** solves it by Gauss-Newton steps with $A = \sum_h \omega_h G_h' \Sigma^{-1} G_h$, Σ updated at every step. The Jacobian G_h is analytic: every parameter enters f directly, through ℓ or through $\ln(bc)$, whose derivatives are immediate, and the restrictions are linear. It agrees with a numerical derivative to 3×10^{-10} (relative).

3.2 Convergence

equaids stops when the Newton decrement $g'A^{-1}g$, where g is the left-hand side of (2), falls below 10^{-6} and Σ is stable to the same relative order. The criterion has a statistical meaning: $g'A^{-1}g$ is the squared distance to the optimum in the metric of the information, so the estimate is then within $\sqrt{10^{-6}} = 0.001$ standard errors of the optimum, in all directions jointly. On the data of Poi and on the Mexican data used below, the distance measured against a solution at 10^{-10} is 4×10^{-5} and 2.4×10^{-4} standard errors.

The iterations stop without convergence when neither the log likelihood nor the scaled gradient has progressed over 20 iterations (steps cut to nothing at a boundary, a singular information matrix). Slow progress is not a stall: Gauss–Newton converges only linearly when the residuals are large, and near the optimum the gradient may fall by only a few percent per iteration. The estimate is started from the mean shares, or from `from()`; five different starting points reach the same optimum on the data of Poi.

4 Elasticities

For a household, with f_i its predicted share,

$$\mu_i = \frac{\partial f_i}{\partial \ln x} = \beta_i + \eta_i' z + \frac{2\lambda_i \ell}{b c}, \quad \mu_{ij} = \frac{\partial f_i}{\partial \ln p_j} = \gamma_{ij} - \mu_i \pi_j - \frac{\lambda_i B_j \ell^2}{b c},$$

with $\pi_j = \alpha_j + \sum_k \gamma_{jk} \ln p_k$ and $B_j = \beta_j + \eta_j' z$. The expenditure, uncompensated and compensated elasticities are

$$e_i = 1 + \frac{\mu_i}{f_i}, \quad e_{ij} = -\delta_{ij} + \frac{\mu_{ij}}{f_i}, \quad e_{ij}^* = e_{ij} + e_i f_j.$$

4.1 Types of elasticities

Definition 1 (Households, individuals, market). With a weight v_h for household h and $S_i = \sum_h v_h f_{ih}$,

$$E_i = 1 + \frac{\sum_h v_h \mu_{ih}}{S_i}, \quad E_{ij} = -\delta_{ij} + \frac{\sum_h v_h \mu_{ijh}}{S_i}, \quad E_{ij}^* = E_{ij} + \frac{\sum_h v_h (f_{ih} + \mu_{ih}) f_{jh}}{S_i},$$

with $v_h = \omega_h$ for the *households*, $v_h = \omega_h n_h$ for the *individuals* (n_h the size of the household, and $\omega_h n_h$ the weight of the whole estimation) and $v_h = \omega_h x_h$ for the *market*.

Since $f_{ih} e_{ih} = f_{ih} + \mu_{ih}$, each of these is the mean of the elasticities of the households, each at its own prices, expenditure and demographics, weighted by $v_h f_{ih}$. No household is singled out as a reference, and there is no point of evaluation: the question whether to take the mean of the log prices or the log of the mean prices does not arise. With $v_h = \omega_h x_h$ the weight is the expenditure of the household on the good, and E_i is the elasticity of the total demand of the population with respect to a proportional change of every household's expenditure (E_{ij} with respect to the price of good j): what a simulation of a price change on aggregate demand, or on tax revenue, needs. None divides by an individual share, and each is a ratio of weighted totals, whose inference under a survey design is standard. All three satisfy the aggregation identities exactly: with $s_i = S_i / \sum_k S_k$, Engel ($\sum_i s_i E_i = 1$), Cournot ($\sum_i s_i E_{ij} = -s_j$), homogeneity ($\sum_j E_{ij} + E_i = 0$) and the symmetry of the compensated matrix ($s_i E_{ij}^* = s_j E_{ji}^*$), since each holds for every household; `equaids` checks them (to about 10^{-15}) and reports them with `checks`. The default is `households`, or `individuals` when `hhsize()` gives the size of the household, as in `easi` and `duvm`; `individuals` is an estimation, not a display. With household data the behaviour is that of the household in all the types: `individuals` gives the elasticities of the household of the average person, not those of a person within the household.

Two other types are available:

- **reference**: the elasticities of a reference household, at the weighted means of $\ln p$, $\ln x$ and z , as [Poi \(2012\)](#). They depend on the point chosen and, the model being nonlinear, differ from the mean of the households;
- **hhmean**: the household elasticities averaged with the sampling weights only, as `demandsys` (which does not weight the average).

The earlier names remain accepted: **aggregate** for **market**, **means** for **reference**, **household** for **hhmean**. Versions 1.0.0 and 1.1.0 reported the market elasticities by default, and called **households** and **individuals** the reference household (with the weights ω_h and $\omega_h n_h$). All five have standard errors (Section 5).

Remark 1 (Weighted by the shares, not a plain mean). On the Mexican cereal data of Section 8.3 (QUAIDS, demographics **hhsz** and **isMale**, sampling weights), with the default $\alpha_0 = 6.31$, the market expenditure elasticity of corn is -0.130 (robust standard error 0.063; 0.064 with the survey design, Section 8.3) and the mean of the household elasticities 0.36; for rice, 0.44 and 1.49. Set $\alpha_0 = 9.5$ instead, an equally admissible value for the fit: the market elasticity of corn becomes -0.085 , a change of less than one standard error, and the mean of the household elasticities -12.19 . The mean moves because some households have a predicted share below 0.001 (225 for corn, up to 1,683 for a good), and their elasticities, which divide by that share, drive it; the weighted means (households, individuals, market) divide by no individual share. **equaids** warns when **elasticities(hhmean)** meets such shares, and the standard errors of the mean show it: 16.9 for corn and 122 for rice, against 0.06 and 0.05 for the market elasticities and 0.017 and 0.033 for those of the households.

The five types on the same data (sampling weights, robust standard errors), for the expenditure and own-price elasticities; the script **replication/types_elasticities.do** reproduces them:

	Households	Individuals	Market	Reference	Mean
<i>Expenditure</i>					
Corn	0.413 (0.017)	0.462 (0.020)	-0.130 (0.063)	0.448 (0.021)	0.356 (16.9)
Wheat	0.573 (0.016)	0.575 (0.018)	0.411 (0.025)	0.584 (0.019)	0.554 (7.17)
Rice	0.659 (0.033)	0.632 (0.038)	0.441 (0.049)	0.681 (0.035)	1.493 (121.8)
Other	1.377 (0.040)	1.321 (0.045)	1.225 (0.040)	1.378 (0.039)	1.298 (23.9)
<i>Own price</i>					
Corn	-0.916 (0.021)	-0.904 (0.025)	-0.848 (0.039)	-0.920 (0.020)	-0.909 (2.16)
Wheat	-0.862 (0.029)	-0.882 (0.032)	-0.809 (0.040)	-0.866 (0.028)	-0.856 (2.31)
Rice	-1.199 (0.067)	-1.148 (0.072)	-1.287 (0.099)	-1.190 (0.064)	-0.882 (46.1)
Other	-0.834 (0.056)	-0.803 (0.060)	-0.860 (0.047)	-0.839 (0.054)	-0.848 (5.04)

The market elasticities differ most: total spending is dominated by richer households, whose shares of the staples are small, and corn becomes an inferior good in the aggregate. The households and the individuals differ little on these data, and the reference household is close to the mean of the households, which it approximates without being it (corn: 0.448 against 0.413); the unweighted mean of the household elasticities is not estimable with any precision.

The at-the-means elasticities of **demandsys** differ from Poi's because **demandsys** evaluates them at the means of the levels of prices and expenditure, Poi and **equaids** at the means of their logarithms; at the same point the formulas agree.

5 Variance

5.1 Influence functions

Every variance in `equaids` is built from influence functions, one row per household: the derivative of the whole procedure with respect to the weight of that household. The iterated FGNLS estimate is the joint solution in (θ, Σ) of

$$\sum_h \omega_h \chi_h = 0, \quad \chi_h = \begin{pmatrix} G_h' \Sigma^{-1} u_h \\ \text{vech}(u_h u_h' - \Sigma) \end{pmatrix},$$

so that

$$\text{IF}_{(\theta, \Sigma), h} = -H^{-1} \omega_h \chi_h, \quad H = \frac{\partial \sum_h \omega_h \chi_h}{\partial(\theta, \text{vech } \Sigma)'},$$

with H the *observed* Jacobian. Its block in θ is $R - A$, where $A = \sum_h \omega_h G_h' \Sigma^{-1} G_h$ is the Gauss–Newton information and $R = \sum_h \omega_h \sum_i (\Sigma^{-1} u_h)_i \partial^2 f_{hi} / \partial \theta \partial \theta'$ the residuals times the second derivatives of the shares; its blocks in Σ carry the estimation of Σ . Both vanish in expectation when the model is right, and the usual sandwich $A^{-1} (\sum_h \omega_h^2 G_h' \Sigma^{-1} u_h u_h' \Sigma^{-1} G_h) A^{-1}$ leaves them out; but they are not zero in a sample, and not at all on real data, where no model is exactly right. The brute force of Section 9.2 found the Gauss–Newton standard errors up to 7.6% too small on the data of [Poi \(2012\)](#), and so are those of `demandsys`, which uses the same information. R is analytic: $\ln a(p)$ and $\ln(b(p)c(p, z))$ are linear in the parameters, so the curvature comes only from the products $B_i \ell$ and $\lambda_i q$ and from $\ln m_0(z)$, and it reduces to weighted cross-products of $\partial \ell / \partial \theta$ and $\partial \ln(bc) / \partial \theta$; computing it costs nothing measurable. An elasticity of the households, the individuals or the market is a ratio $E = T/S$ of totals weighted by v_h (Section 4.1) of functions of θ ; its influence function has two terms,

$$\text{IF}_{E, h} = \underbrace{\frac{v_h (t_h - E s_h)}{S}}_{\text{sampling of the households}} + \underbrace{J \text{IF}_{\theta, h}}_{\text{estimation of } \theta}, \quad J = \frac{1}{S} \left(\frac{\partial T}{\partial \theta'} - E \frac{\partial S}{\partial \theta'} \right),$$

where t_h and s_h are the household contributions to T and S . The first term is what earlier programs leave out: the population over which the elasticity is summarized is itself a sample. The Jacobian J is analytic (the derivatives reduce to weighted column sums of $\partial \ell / \partial \theta$ and $\partial \ln(bc) / \partial \theta$); it agrees with a numerical derivative (central differences with Richardson extrapolation) to 10^{-8} .

For the reference household (`reference`), $E(\theta, \bar{m})$ with $\bar{m}$ the weighted means of $\ln p$, $\ln x$ and z , the sampling term is that of the means, $\text{IF}_{E, h} = J_\theta \text{IF}_{\theta, h} + J_m \omega_h (m_h - \bar{m}) / \sum_h \omega_h$; for the mean of the household elasticities (`hhmean`), $\text{IF}_{E, h} = \omega_h (e_h - E) / \sum_h \omega_h + J_\theta \text{IF}_{\theta, h}$. The derivatives J_θ and J_m are taken by central differences (step $10^{-5} \max(1, |v|)$, an error of the order of 10^{-10}). On the data of [Poi \(2012\)](#), the standard errors of these elasticities are within the spread of a bootstrap of 100 replications of the whole procedure (ratios 0.84 to 1.08).

5.2 Aggregation

The same influence functions give every variance:

- `vce(robust)`, the default: $\frac{N}{N-1} \sum_h \text{IF}_h \text{IF}_h'$;
- `vce(cluster c)`: $\frac{G}{G-1} \sum_g z_g z_g'$, with z_g the sum of the influence functions over cluster g ;
- `vce(svy)`: $\sum_s (1 - f_s) \frac{n_s}{n_s - 1} \sum_i (z_{si} - \bar{z}_s)(z_{si} - \bar{z}_s)'$, with z_{si} the sum over primary sampling unit i of stratum s , n_s the number of units in the stratum and f_s its sampling fraction (first-stage linearization, as Stata's `svy`); the design is read from `svyset`, and the tests use the design degrees of freedom;

- `vce(conventional)`: A^{-1} for the coefficients.

Strata with a single sampling unit follow `svyset`'s `singleunit()`: `certainly`, `scaled` and `centered` as in Stata, and, with the default `missing`, the model is estimated and the standard errors are missing, as Stata does, with a note.

6 Missing prices and non-buyers

On household surveys the prices are often unit values, and a household that does not buy a good has a zero share and no price. Two problems follow: the household leaves the sample for the missing price, and the model of the share ignores that a zero is a decision not to buy. Version 1.1 addresses both.

6.1 Missing prices

With `pimpute(varlist)`, a missing log price is replaced by the weighted mean of the log prices of the households of the sample in the same group of the first variable (the sampling unit, say); when nobody in the group has a price, the group of the next variable is used (urban or rural, say), and so on. Households still without a price leave the sample, and the means are computed again until the sample no longer changes, so that the donors are the households of the final sample and `estat engel` finds the same prices on `e(sample)`. The standard errors include the imputation. A filled price is $m_{gj} = \sum_{d \in D_{gj}} \omega_d \ln p_{dj} / W_{gj}$, the weighted mean over the donors D_{gj} of its group (W_{gj} their weight), so that the weight of a donor moves it by $\omega_d (\ln p_{dj} - m_{gj}) / W_{gj}$; the influence function of the donor gains that change times the derivative, with respect to the filled prices of its group, of every estimating equation (system, Σ , probits) and of the sample terms of the elasticities. The term sums to zero within each group, so it cancels under `vce(cluster)` or `vce(svy)` at the level of the first grouping variable, where only the prices filled at a wider level count; under `vce(robust)` it moved the standard errors of the Mexican estimation by at most 1.7%, 2.5% with the correction for the non-buyers (`replication/pimpute_effect.do`). `vce(bootstrap)` fills the prices again on every replication. On the Mexican data, `pimpute(psu rururb)` keeps 15,674 of the 15,866 households, instead of 10,804 without it: the prices of rice and of the other cereals are filled for 2,502 and 3,508 households, mostly from the urban or rural mean, when no household of the sampling unit buys them.

6.2 The correction of Shonkwiler and Yen

With `selection`, `equaids` applies the two-step method of [Shonkwiler and Yen \(1999\)](#). For a corrected good i , purchase follows a probit, $d_{ih} = 1(s_{ih}'a_i + v_{ih} > 0)$, and the share of a buyer is the share of the model plus an error correlated with v_{ih} . The expected share of a household, buyer or not, is then

$$E[w_{ih}] = \Phi(s_{ih}'a_i) f_{ih}(\theta) + \delta_i \phi(s_{ih}'a_i),$$

with f_i the share of the model (AIDS or QUAIDS), Φ and ϕ the normal distribution and density, and δ_i the covariance of the two errors ($\delta_i = 0$ is no selection). The probits are estimated first, good by good, with the weights of the estimation; the system is then estimated by the same iterated FGNLS on all the households, with the expected shares and the δ_i as further parameters. The last good closes the system, $E[w_M] = 1 - \sum_{i < M} E[w_i]$: it must be bought by every household (the composite of the other goods, in practice) and is not corrected. The options `selgoods()` and `selvars()` are those of `duvm`: the goods corrected (by default, those with zeros) and the variables of the probits only.

Two choices keep the demand theory of the model. The probit is written in prices relative to expenditure, $s_{ih} = (1, \ln(p_{1h}/x_h), \dots, \ln(p_{Mh}/x_h), z_h, q_{ih})$, with q_i the variables of `selvars()`

for good i : it is homogeneous of degree zero, and so are the expected shares. The elasticities are those of the expected demand of all the households,

$$\frac{\partial E[w_i]}{\partial \ln p_j} = \Phi_i \frac{\partial f_i}{\partial \ln p_j} + a_{i,p_j} \phi_i (f_i - \delta_i s_i' a_i), \quad \frac{\partial E[w_i]}{\partial \ln x} = \Phi_i \frac{\partial f_i}{\partial \ln x} - \left(\sum_j a_{i,p_j} \right) \phi_i (f_i - \delta_i s_i' a_i),$$

the first term the change of the share of the buyers (intensive margin), the second the change of the probability of buying (extensive margin). The aggregate elasticities then keep Engel and Cournot aggregation and homogeneity exactly (checked to 10^{-14} on the Mexican data); symmetry holds for the shares f of the model, not for the expected shares.

6.3 Variance

The influence function of the probit coefficients uses the observed Hessian,

$$\text{IF}_{a_i,h} = \omega_h \lambda_{ih} s_{ih}' \left[\sum_h \omega_h \lambda_{ih} (\lambda_{ih} + s_{ih}' a_i) s_{ih} s_{ih}' \right]^{-1}, \quad \lambda = \frac{q \phi(q s' a)}{\Phi(q s' a)}, \quad q = 2d - 1,$$

the derivative of the score the estimator solves; the expected (Fisher) information of the scoring iterations equals it only in expectation, and differed by up to 2% of the standard errors of the probits against the brute force of Section 9.2. The coefficients a enter the system through Φ_i and ϕ_i , and their influence is stacked into that of (θ, Σ) (Section 5, δ in θ):

$$\text{IF}_{(\theta,\Sigma),h} = -H^{-1}(\omega_h \chi_h + H_a \text{IF}_{a,h}'), \quad H_a = \frac{\partial \sum_h \omega_h \chi_h}{\partial a'},$$

with H_a analytic ($\partial u_i / \partial a_i = -\phi_i (f_i - \delta_i s_i' a_i) s_i$, $\partial G_i / \partial a_i = \phi_i s_i \otimes \partial f_i / \partial \theta$). With the Gauss–Newton information in place of H the standard errors of the system were up to 9% too small on the data of the Monte Carlo below (1,000 households, ρ). The elasticities are functions of $\psi = (\theta, a)$: their influence functions add $J_\psi \text{IF}_{\psi,h}$ to the sampling term, J_ψ by central differences, and the variances are aggregated as in Section 5 (robust, cluster, design). `estat engel` draws the Engel curves of the expected shares, with a band from the joint variance of ψ (the bootstrap one under `vce(bootstrap)`).

A table under the header reports, for each corrected good, the percentage of buyers, the pseudo- R^2 of its probit, the households predicted with probability 0 or 1, the variance inflation of the expenditure coefficient β_i due to the selection term, $1/(1 - \rho^2)$ with ρ the correlation of ϕ_i and of the column of β_i in the gradient of the share given the other columns (as the variance inflation of the quality elasticity in `duvm`), and δ_i with its z . Above 10, ϕ_i is almost collinear with $\ln x$ and the correction rests on the curvature of the probit; the good can then be left uncorrected, or a variable of the probit only added.

6.4 Validation

The code was checked against exact oracles: the analytic Jacobian of the expected shares against central differences (3×10^{-11}), the probits against Stata’s `probit` (6×10^{-7}), the correction of the probits in the influence functions against the exact derivative of the estimating equations, and `vce(bootstrap)` against Stata’s `bootstrap` prefix on the same draws (10^{-7}).

The variance was then validated with one change of the model at a time, from the simplest (no demographic, no weight) to the full model, each first by light bootstraps of 20, 30 and 100 replications on data generated by the model with every parameter strongly identified, and finally by a Monte Carlo on the same population. That population (generator `replication/dgp_nonbuyers.do`) has three goods, two of them bought by 73% and 53% of the households and corrected, a variable of the probit only for each (pseudo- R^2 about 0.29; `replication/dgp_describe.do`), zero shares, no price for the non-buyers (filled by `pimpute()`)

from groups of 25 households that share a price shock), and a count demographic (1 to 10) in Ray’s scaling with $\rho = 0.3$. On 700 samples of 3,000 households, every estimation converged, ρ and δ are unbiased (within 1.2 Monte Carlo standard errors), and the analytic standard errors (clustered by the price groups, as the population is) are 0.95 to 1.02 times the standard deviation of the estimates, each within the 95% band of the ratio (`replication/mc_nonbuyers.do`):

	ρ	δ_1	δ_2	$E_{x,1}$	$E_{x,2}$	$E_{x,3}$	E_{11}	E_{22}	E_{33}
s.e. / s.d.	1.023	0.963	0.966	1.021	0.986	0.961	0.956	0.981	0.953
95% band	± 0.054	± 0.049	± 0.055	± 0.052	± 0.053	± 0.053	± 0.051	± 0.052	± 0.051

The band allows for the kurtosis of the estimates in the sampling error of a standard deviation. The slight tilt below one is the usual finite-sample tilt of a sandwich variance. The prices are filled from the price groups, which are also the clusters: the influence of the imputation cancels within them (Section 6.1), and this Monte Carlo checks the rest of the variance; the brute force of Section 9.2 checks the imputation.

6.5 When the linearized standard errors fail

The analytic standard errors rest on a linearization: every parameter well identified and the estimate inside its domain. Two situations of survey data break it, and in both the bootstrap of the whole procedure can give standard errors several times larger than the analytic ones.

The first is a weak selection. When the probits barely separate buyers from non-buyers (a low pseudo- R^2) and no variable of the probit only is available, ϕ_i varies little and $\delta_i \phi_i$ is almost collinear with the rest of the share: the information matrix is nearly singular, and δ_i , and through it the elasticities, move along a ridge from one draw to the next.

The second is Ray’s scaling near its boundary. The correction moves part of the effect of a demographic from the share to the probability of buying; with a count such as the household size, $\hat{\rho}$ can then approach the value at which $m_0(z) = 1 + \rho'z$ reaches zero for the largest households. The estimator is then not regular, and the draws of a bootstrap pile against the boundary. `equaids` measures this distance in standard errors, $\min_h m_0(z_h)/\text{s.e.}(m_0(z_h))$, stored in `e(m0_t)`, and warns below 3. The coefficient ρ itself is often weakly identified, since it enters the shares only through $\beta_i \ln m_0$: its standard error should not be read too literally, while the elasticities are less affected.

In both situations, `vce(bootstrap)` gives the standard errors: it resamples the households (or the sampling units within strata, with `svy`), keeps their weights, and on each replication fills the prices, estimates the probits and the system again; a replication that does not converge is counted and dropped.

6.6 The Mexican cereals

Table below compares the aggregate elasticities on the Mexican data, with the prices of the non-buyers filled by `pimpute(psu rururb)`, without and with the correction (`selection selvars(perc_ocupa)`, the share of employed members in the probits only). Sampling weights. Without the correction the standard errors are the robust analytic ones; with it, the estimate lies 0.5 standard errors from the boundary of Ray’s scaling (the household size is a count, `e(m0_t) = 0.5` against 5.1 without the correction), `equaids` warns, and the standard errors are those of `vce(bootstrap)` (100 replications of the whole procedure); replication in `replication/selection.do`.

Aggregate elasticities of the cereals, without and with the correction for the non-buyers

Good	Expenditure		Expenditure, corrected		Own price		Own price, corrected	
	est.	s.e.	est.	s.e.	est.	s.e.	est.	s.e.
Corn	-0.168	0.039	-0.057	0.033	-0.873	0.030	-0.993	0.029
Wheat	0.341	0.022	0.327	0.036	-0.718	0.034	-0.760	0.030
Rice	0.270	0.043	0.041	0.070	-1.292	0.109	-1.358	0.123
Other	1.148	0.036	0.985	0.046	-0.794	0.058	-0.448	0.110

Good	Buyers (%)	Pseudo- R^2	VIF of β	δ	s.e.
Corn	86.3	0.042	1.1	-0.0795	0.0444
Wheat	81.7	0.016	1.2	-0.0576	0.0322
Rice	29.0	0.030	1.3	-0.0018	0.0041
Other	15.3	0.040	3.3	-0.0020	0.0052

The effect of the correction is not read from δ alone. For corn and wheat, bought by most households, δ is negative, though not significant at 5% once its uncertainty is measured by the bootstrap (the linearized standard errors, 1.3 and 2.2 times smaller, would have made it look clear: $z = -2.4$ and -3.9 instead of -1.8); the expenditure elasticity of corn moves from -0.17 to -0.06 . For rice and the other cereals, bought by 29% and 15% of the households, δ is small, yet the expenditure elasticity of rice falls from 0.27 to 0.04 and the own-price elasticity of the other cereals from -0.79 to -0.45 : without the correction, the model of the share must describe, with one functional form, both whether households buy and how much, and the zeros bend its coefficients; with it, the purchase is modeled by the probit (the extensive margin) and the share of the model describes the buyers.

7 Diagnostics

7.1 At estimation

Before estimating, **equaids** refuses what is not identified (relative prices collinear with each other, log expenditure constant, demographics collinear) and notes what weakens the estimation: near collinearity of the regressors (condition number above 30, [Belsley et al., 1980](#), with variance inflation factors), weak variation of relative prices, demographic variables with a rare modality (effective size $Np(1-p)$) or few integer values, zero shares (not modeled without **selection**), extreme prices. After estimating, it reports why an estimate did not converge (a stall or the iteration limit, with the likely cause: α_0 above the deflated expenditure of most households, steps cut at the boundary $m_0(z) > 0$, a nearly singular information matrix) and notes an optimum near the boundary of Ray's scaling, $m_0(z)$ large for every household (the level of ρ then competes with α_0), households whose expenditure is below the estimated cost $a(p)$, and predicted shares outside $[0, 1]$.

These diagnostics were built on cases that fail. On the data of [Lecocq and Robin \(2015\)](#) (25,776 households, seven goods, demographics **nbpers** and **rural**), a QUAIDS with $\alpha_0 = 10$ has α_0 above the log expenditure of every household; ρ then grows without bound along a ridge of the likelihood, and [Poi \(2012\)](#)'s command stops after twelve minutes at a point that is not the maximum (log likelihood 155,890.8, against 156,634.1 along the ridge), without a warning. **equaids** explains the failure; with α_0 below the smallest $\ln x$ ($\alpha_0 = 2.3$), all three programs reach the same maximum (execution times in [Section 9.3](#)).

7.2 Before estimation: **equaidsdiag**

A diagnostic that needs the model to converge is silent exactly when it does not. **equaidsdiag** takes the syntax of **equaids** and reports on the data and on the regressors at the starting point

(the mean shares, the Stone index plus α_0), without estimating; `estat diagnostics` repeats the diagnostics of an estimate. Table 2 lists the sections.

Table 2: The sections of `equaidsdiag`

Section	What it reports
Households lost	households of the sample dropped for a missing share, price, expenditure or demographic, variable by variable (with unit values as prices, the non-buyers)
D0 Budget shares	mean and aggregate shares, zeros, shares outside $[0, 1]$, shares that do not add up
D1 Relative prices	standard deviation of each relative log price (below 0.05, weak identification of γ), largest correlation between two of them, extreme prices
D2 Expenditure and α_0	range of $\ln x$, α_0 , households with deflated expenditure $\ell \leq 0$ at the start
D3 Demographics	distinct values, range, rare modalities ($Np(1-p)$), negative values (which bring m_0 near 0), largest correlation with a relative log price and correlation with $\ln x$
D4 Conditioning	condition indexes of Belsley et al. (1980) of the Jacobian at the start, and of the block $(1, \ell, \ell^2)$ alone
D5 Small goods	goods with less than 1% of total expenditure
D6 (option)	the model estimated at several values of α_0
D7 (option)	the model estimated again without each demographic in turn

D4 carries the lesson of α_0 . When α_0 lies far from the log expenditures, ℓ varies little relative to its level and ℓ^2 is almost a linear function of ℓ : the quadratic coefficients are then weakly identified, whatever the fit. The condition index of $(1, \ell, \ell^2)$, with uncentered columns so that the level of ℓ counts, measures it. With the default α_0 it is 23 on the data of [Poi \(2012\)](#), 36 on the Mexican cereals and 31 on the data of [Lecocq and Robin \(2015\)](#); on the first two, the standard errors agree with the bootstrap (Section 9); with Poi's $\alpha_0 = 10$ it is 331, the case where they do not. The section notes an index between 30 and 100 and warns above 100.

7.3 Stability to the demographic variables

Estimates of demand systems sometimes shift when a demographic variable is added. With the option `stability`, `equaidsdiag` estimates the model with all the demographics, then again without each of them in turn, on the same households and at the same α_0 , and reports for every good the change ΔE of the aggregate expenditure elasticity and of the aggregate own-price elasticity. Both estimates are computed on the same sample, so the difference of their influence functions, household by household, is the influence function of ΔE , and gives its standard error without further assumption (a Hausman test would need one of the two estimators efficient, which a weighted estimator on a survey is not). A change is marked when $|z| = |\Delta E|/\text{se}$ exceeds the Bonferroni critical value for the $2M$ changes attached to a demographic, $\Phi^{-1}(1 - 0.05/4M)$.

Table 3 shows the three cases met in practice. On the data of [Lecocq and Robin \(2015\)](#), leaving household size out moves the expenditure elasticity of one good by -0.23 ($z = -18.4$); ρ is well identified ($z = 9.2$) and $m_0(z) \geq 1.33$ for every household, so the change is the bias of omitting a variable that affects the shares and moves with expenditure (correlation 0.30). Omitted-variable bias needs both: z_c , built from the prices already in the model (correlation 0.92) and so without an effect of its own on the shares, moves almost nothing. The correlations of D3 are therefore a pointer, not a test, and a threshold on them would miss the Mexican household size (0.22), which D7 marks eight times. The third case is a pure noise variable, z_n : its estimated ρ is -1.11 ($z = -12.6$) and brings $m_0(z) = 1 + \rho'z$ down to 0.02 for a few households, where $\ln m_0$ is nearly singular and a handful of observations drive the fit. Leaving it out moves the own-price elasticity of the first good by -0.058 . The Gauss–Newton standard

Table 3: Stability of the aggregate elasticities: each demographic left out

Data	Demographic	Marked	Largest $ z $	Corr. with $\ln x$	Corr. with prices
Poi, constructed	z_c (moves with a price)	1 of 8	4.1	−0.00	0.92
	z_n (noise)	0 of 8	2.2	−0.00	0.02
Mexico	hhsiz	8 of 10	6.9	0.22	0.10
	isMale	0 of 10	1.9	0.06	0.03
Lecocq–Robin	nbpers	9 of 14	18.4	0.30	0.19
	rural	7 of 14	10.7	−0.07	0.10

Default α_0 , QUAIDS; Mexico with its sampling weights. “Marked”: changes with $|z|$ above the Bonferroni critical value (2.73, 2.81 and 2.91 for four, five and seven goods). The correlation with prices is the largest absolute correlation with a relative log price. Time: 3, 12 and 203 seconds. Script: `replication/stability.do`.

errors of version 1.1.0 marked that change ($z = -3.1$); the exact influence function (Section 5) gives it a standard error 40% larger ($z = -2.2$), and D7 no longer marks it: near the boundary of Ray’s scaling, the curvature that Gauss–Newton leaves out weighs most. A marked change thus says that the elasticities depend on the variable; the estimate of ρ and the smallest $m_0(z)$ (both in `e()`) say whether it is the variable or the scaling that is at work.

8 The command

8.1 Installation

`equaids` requires Stata 14.2 or later and nothing else. It is installed from its GitHub repository, <https://github.com/aabdd12/equaids>:

```
. net install equaids, from("https://raw.githubusercontent.com/aabdd12/equaids/v1
.2.0") replace
```

It installs `equaids`, `equaidsdiag`, the post-estimation commands (`estat diagnostics`, `estat engel`), the help files (`help equaids`, `help equaidsdiag`) and the dialog box. The tables displayed with `stars` and written by `saveres()` are built by a private copy of `tabstars`, shipped with the package, so that no other module is needed. The address points to release 1.2.0, the version described in this note; replacing `v1.2.0` by `main` installs the latest version. The data of Poi (2012) used in the note are Stata’s (`webuse food`); the Mexican data of Section 8.3 are installed with the package (`sysuse mexico_2014_cereals`), with a sample of 3,000 households of the population of Section 6 (`sysuse equaids_nonbuyers`), on which an example shows the correction for the non-buyers with analytic standard errors. Every example of `help equaids` runs from its links, without losing the data in memory.

8.2 Syntax

```
equaids shares [if] [in] [weight], prices(varlist) | lnprices(varlist)
expenditure(varname) | lnexpenditure(varname)
[anot(#)] noquadratic demographics(varlist) pimpute(varlist)
selection selgoods(namelist) selvars([good:] varlist [; ...])
vce(robust | cluster clustvar | svy | conventional
    | bootstrap [, reps(#) seed(#) strata() psu() svy]) level(#)
elasticities(households | individuals | market | reference | hhmean)
hhsiz(varname)
snames(namelist) dec(#) dislas(#) dregres(#) compensated checks
detail stars noelastse saveres(filename) notable
from(matname) tolerance(#) iterate(#) nolog]
```

Weights are `aweight`, `fweight`, `pweight` and `iweight`; with `vce(svy)` the weight comes from `svyset`. The names of the options are those of `easi` and `duvm`. Users of `wquaid`s replace `hweight(w)` by `[pw=w]`, `model(2)` by `noquadratic` and `xfil()` by `saveres()`. Results are returned in `e()`: coefficients in Poi's parameterization with their variance, the free parameters, Σ , the five types of elasticities and their variances, the diagnostics and the design.

8.3 Example: Mexican cereals with the survey design

The Mexican household survey ENIGH 2014 gives budget shares of four cereals (corn, wheat, rice, other) and a composite of the other goods, with unit-value prices, two demographics (household size and the sex of the head), and the full design: 197 strata, sampling units and weights. The example diagnoses, estimates and draws the Engel curves:

```
svyset psu [pweight=sweight], strata(strata) singleunit(centered)
local S "wcorn wwheat wrice wother wcomp"
local O "prices(pcorn pwheat price pother pcomp) expenditure(hh_current_inc)
        demographics(hhsize isMale)"
equaidsdiag `S' [pw=sweight], `O'
equaids `S', `O' snames(corn wheat rice other composite) vce(svy)
estat engel
```

The weight is the one used for the Mexican results of the other sections, so the point estimates are theirs; the strata and sampling units change only the standard errors.

Diagnostics. `equaidsdiag` reports seven warnings before any estimation. 5,062 of the 15,866 households (32%) are lost, mostly to missing unit values: the non-buyers of rice and of the other cereals have no price. Rice and the other cereals are zero for 68% and 80% of the households and weigh 0.13% and 0.15% of total expenditure (D5, small goods). The condition index of $(1, \ell, \ell^2)$ is 36 with the default $\alpha_0 = 6.31$ (D4, moderate). The demographics do not move with prices (largest correlation 0.10), and household size moves moderately with expenditure (0.22); D7 marks it (Table 3).

Estimation. On the 10,804 remaining households (1,464 sampling units, 1,267 design degrees of freedom), the QUAIDS converges in 19 iterations and 3 seconds. Ten strata keep a single sampling unit after the households with missing prices are dropped; the example centers them (`singleunit(centered)` in `svyset`); with Stata's default, `missing`, the standard errors would be missing. Table 4 gives the aggregate elasticities.

Table 4: Mexican cereals: aggregate elasticities, design and robust standard errors

Good	Share (%)	Expenditure			Own price		
		E_i	se design	se robust	E_{ii}	se design	se robust
Corn	1.42	-0.130	0.064	0.063	-0.848	0.066	0.039
Wheat	1.18	0.411	0.026	0.025	-0.809	0.057	0.040
Rice	0.12	0.441	0.046	0.049	-1.287	0.102	0.099
Other	0.16	1.225	0.054	0.040	-0.860	0.043	0.047
Composite	97.12	1.024	0.001	0.001	-1.016	0.002	0.001

QUAIDS, demographics `hhsize` and `isMale`, $\alpha_0 = 6.31$, 10,804 households. Share: predicted aggregate budget share. Design: `vce(svy)`, strata, sampling units and weights; robust: `vce(robust)` with the same weights (`replication/example_design.do`).

Corn is an inferior good in the aggregate ($E = -0.13$), wheat and rice are necessities, the other cereals a luxury. The design matters most for the price elasticities: the design standard error of the own-price elasticity of corn is 1.7 times the robust one, of wheat 1.4 times; for the expenditure elasticities the two are close, except for the other cereals (1.3) and the composite (1.5, on a standard error of 0.001).

Engel curves, QUAIDS (equaids)

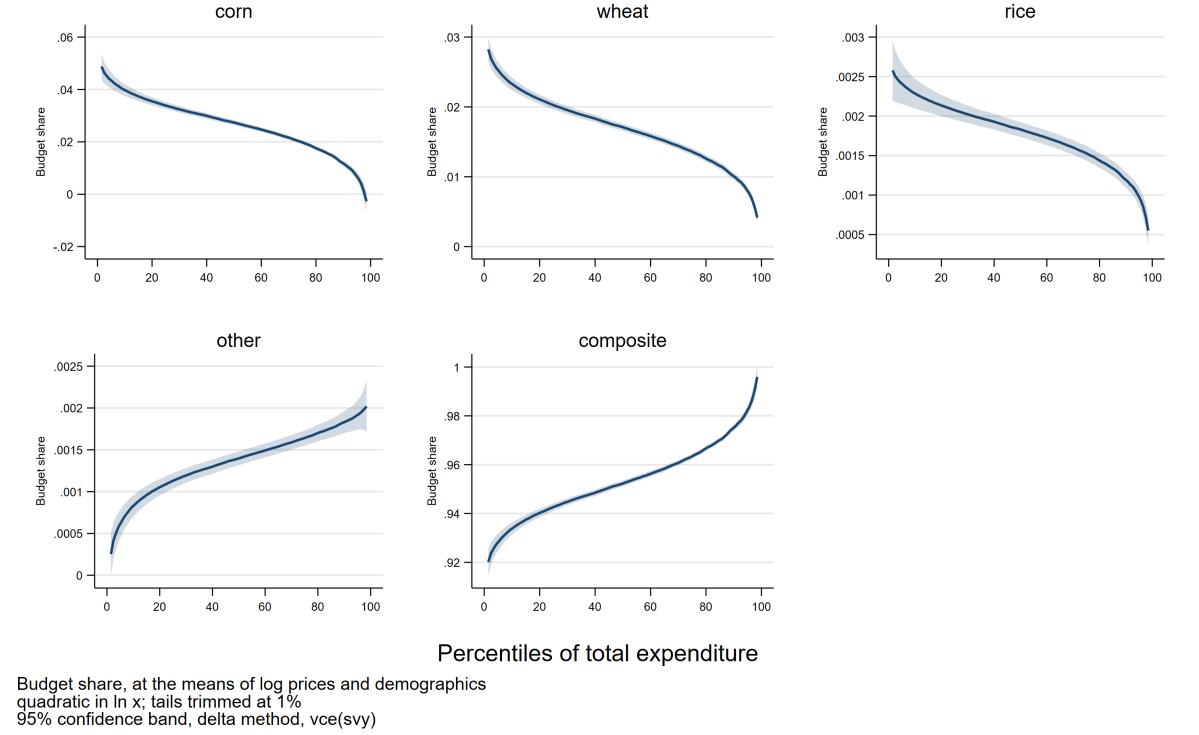


Figure 1: Mexican cereals: Engel curves at the means of log prices and demographics (`estat engel`), with 95% design-based bands

Engel curves. Figure 1 shows the curves at the means of the log prices and demographics. The quadratic term appears as curvature rather than as a turning point: no curve turns inside the plotted range, the 1st to the 99th percentile of expenditure (the turning points, in `r(turn)`, lie below the poorest household for corn, rice and the composite, at the 99.5th percentile for the other cereals, and far above the richest household for wheat). The curvature bends the share of corn below zero from the 98.5th percentile of expenditure on (the 95% band includes zero from the 97.5th): QUAIDS extrapolates the fall of a small, inferior good beyond zero among the richest households; `equaids` notes 264 households with a predicted share of some good outside $[0, 1]$. This is the case of the small goods of Section 10: the aggregate elasticity is unaffected, since it weighs each household by its expenditure on corn, but household-level predictions at the top of the distribution should not be read. With `asobserved observed`, `estat engel` adds the smoothed observed shares to judge the fit of the functional form along expenditure.

9 Validation

Besides the results reported here, the module was developed against 293 automated checks: equality with earlier programs, analytic derivatives against numerical ones, the aggregation identities, exact oracles for the variance, and each diagnostic built on a constructed case.

9.1 Against earlier programs and exact oracles

Under the same model and α_0 , `equaids` reproduces the estimates of Poi (2012) and `demandsys` on the data of Poi (2012) (4,048 households, four goods), on Mexican cereal data (ENIGH 2014, 10,804 households, five goods) and on the data of Lecocq and Robin (2015): log likelihoods to

about 10^{-14} and coefficients to the precision of their convergence (10^{-7} to 2×10^{-6}), AIDS and QUAIDS, with and without demographics and weights. The robust and cluster variances of **demandsys** use the Gauss–Newton information; **equaids** uses the observed Jacobian (Section 5), and its robust standard errors are up to 8% larger than **demandsys**’s on the data of Poi (2012). The brute force below says which are right.

For the survey design there is an exact oracle. Dividing each influence function by the sampling weight and giving the result to Stata’s **svy: total** makes Stata’s own linearization compute the design variance of $\sum_h \text{IF}_h$, which is what **equaids** reports. The two agree to 4×10^{-15} (relative) for strata and sampling units, the finite-population correction (as a rate and as a population size) and the four treatments of single-unit strata.

9.2 The influence function by brute force

A bootstrap scores a standard error against another estimate of it, with its own noise (about 3% at 500 replications, 7% at 100): it cannot see a term of the variance that weighs a few percent. The influence function can be computed without noise, by brute force. For each household h in turn, the *whole* procedure is run again with its weight at $1 \pm \varepsilon$ ($\varepsilon = 10^{-2}$, **iweights**): the imputation of the prices, the probits, the system and Σ , the elasticities. $U_h = \partial \hat{\theta} / \partial w_h$ by central differences is the influence function of every estimate, and $\sqrt{n/(n-1) \sum_h U_h^2}$ the robust standard error in which nothing is neglected (the infinitesimal jackknife: for a mean, $U_h = (x_h - \bar{x})/n$ and it is $s/\sqrt{n}$). The households run in blocks in parallel processes; the 8,096 estimations of the data of Poi (2012) take 8 minutes on four processes.

Table 5: Reported robust standard error divided by the brute-force one: range over the estimates

Data	Estimates	Gauss–Newton	equaids
Poi (2012), 4,048 households, QUAIDS	166	0.924–1.002	1.0000
Non-buyers, 1,000 households, prices observed	126	0.909–1.022	1.0000
Non-buyers, prices filled by pimpute()	126	0.911–1.020	0.9998–1.0000

Estimates: the coefficients (with δ and the probits under the correction for the non-buyers) and the expenditure, uncompensated and compensated elasticities of the households, the market, the reference household and the household mean (Gauss–Newton: all but the households). Gauss–Newton: A^{-1} in the influence function of the coefficients, Σ held fixed, the expected information of the probits, the filled prices taken as data. Non-buyers: the design of Section 6 (**dgp_nonbuyers.do**), correctly specified. Largest deviation of **equaids**: 2×10^{-4} , the precision of the brute force. Scripts: **replication/bruteforce/**.

Table 5 gives the result. Adding the pieces one at a time to the influence function locates each gap: on the data of Poi (2012) the observed Hessian with Σ held fixed brings the coefficients from 0.924–1.000 to 0.978–1.001, and the estimation of Σ closes the rest (the price coefficients γ); under the correction for the non-buyers the observed Hessian of the probits closes their block (up to 2%), not the system’s (up to 9%, ρ), which needs the stacked Jacobian; the imputation of the prices weighs up to 0.4% there, up to 2.5% on the Mexican data. The gaps appear even on a correctly specified model, and they would be invisible to a bootstrap of a few hundred draws; on the Mexican data the bootstrap confirms them independently: the ratios of the coefficients to their bootstrap standard deviations went from 0.79–1.29 to 0.93–1.05 (Table 7).

9.3 Estimation performance

The three programs reach the same estimates; they do not take the same time. Table 6 gives the execution times on the three data sets, for the same model and α_0 , without weights. The

three programs reach the same log likelihood (to 4×10^{-10}). `equaids` is 15 to 21 times faster. The difference is the Jacobian: `nlsur`, on which `Poi (2012)` and `demandsys` rest, differentiates the model numerically, one evaluation of the whole system per parameter and per iteration (53 parameters for the QUAIDS on the data of `Lecocq and Robin (2015)`), where `equaids` computes the analytic Jacobian in one pass. What remains long in `equaids` on these data is the number of iterations of the AIDS (88): Gauss–Newton converges only linearly when the residuals are large (Section 3.2).

Table 6: Execution times (seconds), same model and α_0 , no weights

Data, model	Households	Goods	Parameters	equaids		Poi	demandsys
				default	10^{-10}		
<code>Poi (2012)</code> , QUAIDS	4,048	4	15	0.2	0.2	4.2	3.7
Mexico, QUAIDS, demographics	10,804	5	32	2.8	3.5	58.2	49.3
<code>Lecocq and Robin (2015)</code> , QUAIDS	25,776	7	53	50	66	839	779
<code>Lecocq and Robin (2015)</code> , AIDS	25,776	7	47	86	108	1,637	1,482

One Windows computer, Stata 19. α_0 : 2.2026, 6.3132 and 2.3; demographics `hhsize` and `isMale` (Mexico), `nbpers` and `rural` (`Lecocq and Robin`). `equaids` at its default tolerance and at 10^{-10} ; Poi and `demandsys` at theirs. Poi on the data of `Lecocq and Robin` was timed in a separate run. The times of `equaids` are those of version 1.0.0, which reported the market elasticities with their standard errors; later versions reach the same estimates, and their default output adds the standard errors of the other types of elasticities. Script: `replication/timing.do`.

9.4 Against the bootstrap

The exact oracles check the computation; the bootstrap checks that the linearized standard errors are right. Each replication re-estimates the model with the whole procedure, α_0 recomputed by the default rule. Table 7 compares the analytic standard errors of the aggregate elasticities with the bootstrap standard deviations.

Table 7: Analytic standard errors / bootstrap standard deviations

Data and bootstrap	Draws	Elasticities	Ratio range	θ , median	θ , range
Poi’s data, no weights; households	1,000	36	0.96–1.03	1.00	0.98–1.02
Mexico, pweights, demographics; households	500	55	0.93–1.06	1.00	0.93–1.05
Mexico, survey design; Rao–Wu (<code>Rao and Wu, 1988</code>)	500	55	0.95–1.06	1.01	0.94–1.06

All replications converged. The Monte Carlo band of a ratio is $1 \pm 2.5/\sqrt{2B}$ (0.056 for 1,000 draws, 0.079 for 500); no elasticity and no coefficient lies outside it. With the Gauss–Newton information in the influence functions (Section 5), the coefficients ranged from 0.92 to 1.01, 0.79 to 1.29 and 0.77 to 1.21: outside the band on the Mexican data.

The survey design matters: on the Mexican data the design standard errors of the price elasticities are 13 percent larger than the robust ones at the median, and up to 70 percent larger.

With Poi’s $\alpha_0 = 10$ held fixed, on his data, the same comparison fails, and instructively: the ratio for the coefficients is 0.62 at the median (0.31 for λ_1 , whose bootstrap mean is 1.1 standard errors from the estimate), and 0.54 to 0.94 for the expenditure elasticities. A parametric simulation $\theta \sim N(\hat{\theta}, \hat{V})$ with the data held fixed reproduces the delta-method standard deviation of the elasticities, so the formula is not at fault: the estimator itself is far from normal when the information matrix is nearly singular. The robust coefficient standard errors of `Poi (2012)` and `demandsys`, at the same α_0 , which use the Gauss–Newton information, give a ratio of 0.57.

9.5 Monte Carlo

The bootstrap checks the standard errors on real data, where the truth is unknown. A Monte Carlo checks them against a known truth, and for that the data-generating process must be the *perfect case* of the formula: the model correctly specified, consistent with consumer theory, and every parameter clearly identified. Weak identification is not what a variance formula is validated on; it is the business of the diagnostics (Section 7).

Populations. Six finite populations of 100,000 households (AIDS and QUAIDS, three error types), each in 10 strata of 400 primary sampling units of 25 households. Log prices vary by sampling unit (spatial variation) with a small household noise; log expenditure is normal, $\ln x = 5.5 + 0.12(s - 5.5) + v_{\text{psu}} + \varepsilon$ with $v_{\text{psu}} \sim N(0, 0.1^2)$ and $\varepsilon \sim N(0, 0.6^2)$ in stratum s ; two demographics, an indicator ($p = 0.4$) and a household size from 1 to 8. The shares are the model at known parameters (mean shares 40, 30, 20 and 10%) plus errors with standard deviations 0.2 times the mean shares and the correlations of a multinomial: homoskedastic, heteroskedastic (scaled by $\exp((\ln x - \overline{\ln x})/2)$), or clustered (30% of the variance at the level of the sampling unit). Before any draw, the script checks that the Slutsky matrix is symmetric negative semidefinite and the systematic shares in $[0, 1]$ for every household, that the turning points of the quadratic Engel curves lie inside the range of ℓ with the condition index of $(1, \ell, \ell^2)$ below 100, and that every free parameter has $|\theta|/\text{se} \geq 4$ at $n = 5,000$ (4.2 to 5.3 over the six populations). The truth is the census: **equaids** on the whole population.

Samples. Simple random sampling: 5,000 households drawn independently, **vce(robust)**. Survey design: 200 sampling units drawn without replacement within strata, the poorer strata oversampled (32 to 10 units per stratum), all their households, weights the inverse probabilities, **svyset** with strata and the finite-population correction, **vce(svy)** and t with 190 degrees of freedom. 1,000 replications per cell.

Table 8: Monte Carlo: coverage of the 95% intervals and standard error over standard deviation

Model	Errors	Design	Coverage	SE / SD
AIDS	homoskedastic	random	0.949–0.955	1.006–1.023
		survey	0.942–0.946	0.977–0.988
	heteroskedastic	random	0.950–0.953	1.003–1.022
		survey	0.940–0.946	0.970–0.986
	clustered	random	0.952–0.954	1.006–1.011
		survey	0.941–0.943	0.969–0.980
QUAIDS	homoskedastic	random	0.948–0.951	0.988–1.006
		survey	0.945–0.948	0.987–1.011
	heteroskedastic	random	0.945–0.953	0.980–1.008
		survey	0.943–0.948	0.984–1.002
	clustered	random	0.949–0.952	0.998–1.001
		survey	0.932–0.944	0.975–0.986

Range over four families of statistics: aggregate expenditure, uncompensated and compensated elasticities, and the free coefficients. Coverage: mean over the family; SE/SD: median over the family of the mean analytic standard error over the Monte Carlo standard deviation. Monte Carlo bands for 1,000 draws: coverage 0.95 ± 0.017 , ratio 1 ± 0.056 .

Table 8 gives the results. Every replication converged, in 9 to 11 iterations at the median, no estimate lies more than 10 robust standard deviations from the median, and the bias is at most 0.12 Monte Carlo standard deviations. Under random sampling, coverage is 0.945 to 0.955 and the ratio 0.98 to 1.02. Under the survey design, coverage is 0.932 to 0.948 and the ratio 0.97 to 1.01: slightly below, within the bands for all but one family (the expenditure elasticities of the QUAIDS with clustered errors, 0.932). This is the small-sample limit of the linearized design variance, which estimates the variance within each stratum from its sampling units. With 100 units (5 to 16 per stratum) instead of 200, the design coverage was 0.925 to 0.944 with the Gauss–

Newton variance of version 1.1.0, while the random-sampling cells, on the same populations, did not move.

10 Practical advice

The tests of this note, and the cases that failed while building them, give the following advice to the practitioner.

1. α_0 . Keep the default, just below the smallest log expenditure. A value above the log expenditures (Poi's example value of 10) fits as well but makes the quadratic block nearly collinear: the coefficient standard errors are then up to three times too small. To impose a value, check the condition index of $(1, \ell, \ell^2)$ in `equaidsdiag` (below 30 is safe, above 100 is not) and the sensitivity of the elasticities (option `sensitivity`).
2. *The summary elasticity*. Report the aggregate elasticity. The mean of the household elasticities divides by predicted shares that can be near zero; on the Mexican data it changes from 0.36 to -12.19 with α_0 while the aggregate elasticity moves by less than one standard error.
3. *Small goods*. A good below 1% of total expenditure (D5) is poorly served by QUAIDS: predicted shares near zero or negative for some households (even for corn, 1.6% of total expenditure, in Section 8.3), which make the mean of the household elasticities unstable (Section 4). Grouping small goods is often the better choice.
4. *The survey design*. Declare it and use `vce(svy)`: on the Mexican data the design standard errors of the price elasticities are 13% larger than the robust ones at the median, and up to 70% larger. The linearized design variance needs enough sampling units per stratum: with 10 to 32 per stratum the Monte Carlo coverage is 0.932 to 0.948, with 5 to 16 it was 0.925 to 0.944 (with the variance of version 1.1.0). With few units per stratum, read the intervals as slightly narrow, or use a design bootstrap (Rao and Wu, 1988), which `equaids` does not provide. Strata with a single unit give missing standard errors by default, as in Stata.
5. *Demographic variables*. Enter them as counts or indicators, nonnegative, so that $m_0(z) = 1 + \rho'z$ stays away from zero; a categorical variable enters as indicators. Check the estimate of ρ and the smallest $m_0(z)$ in `e()`: a fit where m_0 nears zero for a few households is driven by them.
6. *Stability*. Run `equaidsdiag, stability`. A demographic whose omission moves the elasticities beyond sampling noise affects the shares and moves with expenditure or prices: it belongs in the model, unless its ρ is at work (previous point).
7. *Zeros and missing prices*. With unit values as prices, the non-buyers have none: fill them with `pimpute()` (the sampling unit, then a wider area) rather than lose the households, and correct the goods with many zeros with `selection` (Section 6). Read the selection table: a variance inflation of β above 10 says that the correction rests on the curvature of the probit; a variable that moves the purchase but not the share of the buyers, in `selvars()`, helps. With weak probits, or when `equaids` warns that the estimate is near the boundary of Ray's scaling, use `vce(bootstrap)`: the linearized standard errors can then be several times too small.
8. *Convergence*. The default criterion puts the estimate within 0.001 standard errors of the optimum. When an estimate does not converge, `equaids` says why (a stall, the iteration limit, a boundary of m_0 , a singular information matrix); α_0 above the log expenditures is a common cause (Section 7).

11 Conclusion

equaids estimates the same AIDS and QUAIDS as [Poi \(2012\)](#) and **demandsys**, by the same estimator, and reproduces their estimates. What it adds is for the main use of these systems, household surveys: an aggregate elasticity defined for the population and stable where the mean of the household elasticities is not; standard errors that include the sampling of the households and the survey design, checked against exact oracles, the bootstrap and a Monte Carlo on populations where the truth is known; a default α_0 that keeps the estimator well conditioned; and diagnostics that speak before the estimation, including the stability of the elasticities to each demographic variable; and, for the goods that many households do not buy, the filling of their prices and the correction of [Shonkwiler and Yen \(1999\)](#), with standard errors that include the probits. Instrumental variables for expenditure and prices ([Lecocq and Robin, 2015](#)), tests of homogeneity and symmetry, subpopulations and the EASI system ([Lewbel and Pendakur, 2009](#)) are the next steps.

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